

Fakhar STEM Sphere



**Directorate of public
instructions
(Colleges) Punjab**
**Academic Support for
Targeted Students**
First Year Mathematics

PAIRING SCHEME FOR SMART SYLLABUS**INTERMEDIATE – I**

Sr. No	Chapter Number	MCQ's	Short Questions	Long Questions	Marks
1	1	2	4	1	15
2	2	1	2	Unit 2 or Unit 3	10
3	3	1	2	Unit 2 or Unit 3	10
4	4	2	3	1	13
5	5	1	1	Unit 5 or Unit 9	8
6	6	2	4	1	15
7	7	2	3	1	13
8	9	1	2	Unit 5 or Unit 9	10
9	10	2	3	1	13
10	11	1	2		5
11	12	1	3	1	12
12	13	2	4	1	15
13	14	2	4	1	15
	Grand Total	20	37	50	144

Q.1	MCQS (20 Marks)
Short Questions	
Q.2	Chapters 1+2+3+4+5 (8/12)
Q.3	Chapters 6+7+9+10 (8/12)
Q.4	Chapters 11+12+13+14 (9/13)
Long Questions (Attempt any three)	
Q.5	Chap 1 + Chap 2 or Chap 3
Q.6	Chap 4 + Chap 5 or Chap 9
Q.7	Chap 6 + Chap 7
Q.8	Chap 10 + Chap 12
Q.9	Chap 13 + Chap 14

Important Chapters (For Average Students)

Sr. No	Chapter Number	MCQ's	Short Questions	Long Questions	Marks
1	1	2	4	1	15
2	4	2	3	1	13
3	5	1	1	Unit 5 or Unit 9	8
4	9	1	2	Unit 5 or Unit 9	10
5	12	1	3	1	12
6	13	2	4	1	15
7	14	2	4	1	15

Weightage for Average Students**Skip Chapters 2,3,6,7,10,11**

For Attempted MCQS Marks App.	11
For Attempted Short Questions Marks App.	38
For Attempted Long Questions Marks App.	25
Total	74

MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 1

MCQs

1. The equation $x^2 + 1 = 0$ has the solution in set:
(A) R (B) C (C) Q (D) Q'
2. The numbers of the form $x + iy$, where $x, y \in R$ and $i = \sqrt{-1}$ are called:
(A) Complex numbers (B) Natural numbers
(C) Real numbers (D) Rational number
3. The imaginary part of a complex number $\overline{a + ib}$ is:
(A) $-b$ (B) b (C) a (D) $-a$
4. For any complex number z , $z = \bar{z}$ iff:
(A) $\operatorname{Re}(z) = 1$ (B) $\operatorname{Re}(z) = 0$ (C) $\operatorname{Im}(z) = 1$ (D) $\operatorname{Im}(z) = 0$
5. If $z = -2 + 3i$, then $\bar{z} =$:
(A) $-2 - 3i$ (B) $2 - 3i$ (C) $-2 + 3i$ (D) $2 + 3i$
6. A set of complex numbers is denoted by:
(A) $C = \{a + bi \mid a, b \in R \text{ and } i^2 = -1\}$ (B) $C = \{a + bi \mid a, b \in P \text{ and } i^2 = -1\}$
(C) $C = \{a + bi \mid a, b \in R \text{ and } i^2 = 1\}$ (D) $C = \{a + bi \mid a, b \in R \text{ and } i = -1\}$
7. The real part of $z = i(3 - 4i)$ is:
(A) 4 (B) -4 (C) -5 (D) 5
8. The imaginary part of $\frac{5i - 2}{7}$ is:
(A) 2 (B) -5 (C) $\frac{5}{7}$ (D) $\frac{5}{7}i$
9. $i^2 \cdot i^4 \cdot i^6 \cdot i^8$ equal to:
(A) 1 (B) 0 (C) -1 (D) i
10. $(1 + i)^{10} =$:
(A) $32(1 + i)$ (B) 32 (C) i (D) $32i$

11. $(-i)^{17} =:$

- (A) $-i$ (B) -1 (C) i (D) 1

12. $\sum_{n=1}^{10} i^n =:$

- (A) $1+i$ (B) 0 (C) i (D) $i-1$

13. $(9+5i) - (-8+i) =:$

- (A) $-17-4i$ (B) $1+6i$ (C) $17+4i$ (D) $17-4i$

14. $a+ib$ form of the complex number $(1+2i)(-2+i)$ is:

- (A) $-4-3i$ (B) $4-3i$ (C) $-4+3i$ (D) $4+3i$

15. Which of the following represents the ordered pair form of the complex number $5-2i$?

- (A) $(5, 2)$ (B) $(5, -2)$ (C) $(-5, 2)$ (D) $(-5, -2)$

16. The multiplicative inverse of $(1, -2) =:$

- (A) $\left(\frac{1}{5}, \frac{-2}{5}\right)$ (B) $\left(\frac{-1}{5}, \frac{-2}{5}\right)$ (C) $\left(\frac{1}{5}, \frac{2}{5}\right)$ (D) $\left(\frac{-1}{5}, \frac{2}{5}\right)$

17. Multiplicative inverse of $(a, 0)$ if $a \neq 0$ is:

- (A) $\left(\frac{1}{a}, 0\right)$ (B) $\left(\frac{1}{(a, 0)}\right)$ (C) $(-a, 0)$ (D) $\left(0, \frac{1}{a}\right)$

18. The square root of $7-24i$ is:

- (A) $\pm(3-4i)$ (B) $\pm(4-3i)$ (C) $\pm(4+3i)$ (D) $\pm(3+4i)$

19. The square roots of i are:

- (A) $\pm\left(\frac{1+i}{\sqrt{2}}\right)$ (B) $\frac{1+i}{2}$ (C) $\pm\frac{(1+i)}{2}$ (D) $\frac{1+i}{\sqrt{2}}$

20. If $P(z) = z^3 + 2z^2 + 1$ is a polynomial, how many zeroes does it have in the complex plane?

- (A) 0 (B) 1 (C) 2 (D) 3

21. If $P(z) = z - (3 - 2i)$, what is the root of $P(z) = 0$?
(A) $3 + 2i$ (B) $-3 + 2i$ (C) $3 - 2i$ (D) $-3 - 2i$

22. Find a polynomial $P(z)$ of degree 3 with zeros $1, i, -i$ and satisfying $P(2) = 5$:

- (A) $P(z) = (z - 1)(z^2 + 1)$ (B) $P(z) = 2(z - 1)(z^2 + 5)$
(C) $P(z) = -3(z - 1)(z^2 + 1)$ (D) $P(z) = (z + 1)(z^2 - 1)$

23. Factorized form of $P(z) = 3x^2 + 6y^2$ is:

- (A) $3(x + \sqrt{2}y)(x - \sqrt{2}y)$ (B) $(3x - 6jy)(3x + 6jy)$
(C) $3(x + \sqrt{2}y)^2$ (D) $3(x + \sqrt{2}yi)(x - \sqrt{2}yi)$

24. Cube roots of unity are:

- (A) $-1, -2, 1$ (B) $-1, -1, \omega$ (C) $1, \omega, \omega^2$ (D) $-1, -\omega, \omega^2$

25. If ω is the cube roots of unity then $(1 + \omega - \omega^2)^8 =$:

- (A) 256 (B) -256 (C) -256ω (D) 256ω

26. If ω is a cube root of unity, then conjugate of ω is:

- (A) ω^2 (B) $-\omega^2$ (C) $-\omega$ (D) $-i$

27. $\omega^{-29} =$:

- (A) 0 (B) 1 (C) ω^2 (D) ω

28. $1 + \frac{1}{\omega} + \omega = ?$

- (A) 1 (B) ω (C) 0 (D) ω^2

29. The three cube roots of -64 are:

- (A) $2, 2\omega, 2\omega^2$ (B) $-4, -4\omega, -4\omega^2$ (C) $-2, -2\omega, -2\omega^2$ (D) $4, 4\omega, 4\omega^2$

30. The product of three cube roots of -27 is:

- (A) 27 (B) -27 (C) 1 (D) 0

31. Four fourth roots of unity are:

- (A) $-1, 1, i, -i$ (B) $-2, 2, -2i, 2i$ (C) $-3, 3, 3i, -3i$ (D) $\pm 4, \pm 4i$

32. The four fourth roots of 16 are:

- (A) $2, -2, 2i, -2i$ (B) $1, -1, i, -i$ (C) $3, -3, 3i, -3i$ (D) $4, -4, 4i, -4i$

33. The sum of all fourth roots of 16 is:

- (A) 16 (B) -16 (C) 0 (D) 1

34. The product of all fourth roots of 81 is:

- (A) -1 (B) 0 (C) 81 (D) -81

35. The argument of $z = i$ is:

- (A) 0° (B) 90° (C) 180° (D) 270°

36. Polar form of $z = 4 + 4\sqrt{3}i$ is:

- (A) $8\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ (B) $4\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
(C) $8\left(\cos \frac{2\pi}{3} + i \sin \frac{5\pi}{3}\right)$ (D) $4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$

EXERCISE 1.1

SHORT QUESTIONS

1. Find the multiplicative inverse of each of the following numbers.

- (i) $(\sqrt{2}, -\sqrt{5})$ (ii) $(1, 0)$

2. Separately into real and imaginary parts (write as a simple complex number).

- (i) $\frac{2-7i}{4+5i}$ (ii) $\frac{(4+3i)^2}{4-3i}$

3. Prove that $\bar{\bar{z}} = z$ iff z is real.

4. For $z \in \mathbb{C}$, show that (i) $\frac{z - \bar{z}}{2i} = \text{Im}(z)$ (ii) $|z|^2 = z \cdot \bar{z}$

5. If $z_1 = 2 + 7i$ and $z_2 = -5 + 3i$, then evaluate the following:

(i) $|3z_1 + 2\bar{z}_1|$ (ii) $|(z_1 + z_2)^3|$

6. Show that $i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4} = 0$, for all $n \in \mathbb{N}$.

7. Find the least positive value of n , if $\left(\frac{1+i}{1-i}\right)^{2n} = 1$

EXERCISE 1.2

SHORT QUESTIONS

1. Find the values of x and y if: (i) $(x + iy)^2 = 25 + 60i$ (ii) $(x + iy)^2 = 64 + 48i$

2. If $z_1 = 2 + 3i$ and $z_2 = 1 - \alpha$, find the value of α such that $\text{Im}(z_1 z_2) = 7$

3. Show that $z_1, z_2 \in \mathbb{C}$, $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$

4. Find the square root of the following complex numbers:

(i) $-7 - 24i$ (ii) $119 + 120i$

5. Find the square root of $13 - 20\sqrt{3}i$ and represent them on an Argand diagram.

6. If $z_1 = 4 + 5i$ and $z_2 = \alpha - 2i$, find the value of α such that $\text{Re}(z_1 z_2) = 20$

LONG QUESTIONS

1. If $z_1 = -13 + 24i$ and $z_2 = x + yi$, find the values of x and y such that $z_1 - z_2 = -27 + 15i$

2. Find the values of x and y in each of the following:

(i) $x + iy + 2 - 3i = i(5 - i)(3 + 4i)$

(ii) $\frac{x}{2+i} + \frac{y}{3-i} = 4 + 5i$

EXERCISE 1.3

SHORT QUESTIONS

1. Factorize the following:

(i) $a^2 + 4b^2$

(ii) $9a^2 + 16b^2$

(iii) $z^2 - 2iz - 1$

(iv) $2z^2 - 22z + 65$

2. Factorize the following polynomial into its linear factors:

(i) $z^3 + 8$

(ii) $z^3 - 2z^2 + 16z - 32$

(iii) $z^4 + 21z^2 - 100$

(iv) $z^4 - 16$

(v) $z^4 + 7z^2 - 144$

3. Solve the following equations:

(i) $2z^4 - 32 = 0$

(ii) $5z^5 - 5z = 0$

(iii) $4z^4 - 25z^2 - 21 = 0$

(iv) $z^3 + z^2 + z + 1 = 0$

LONG QUESTIONS

1. Solve the following complex quadratic equation by completing square method:

(i) $2z^2 - 3z + 4 = 0$

(iii) $3z^2 - 18z + 50 = 0$

(iii) $2z^2 + 6z + 9 = 0$

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2. Find a polynomial of degree 3 with zeros $3, -2i, 2i$ and satisfying $P(1) = 20$

3. Find a polynomial of degree 4 with zeros $4, -4, 1+i, 1-i$ and satisfying $P(2) = 72$

EXERCISE 1.4

SHORT QUESTIONS

1. Find the three cube roots of: (i) 8 (ii) -27 (iii) -125

2. Find the four fourth roots of 16, 625. Also show that their sum is zero in each case.

3. Evaluate: (i) $\left(\frac{-1+\sqrt{-3}}{2}\right)^7 + \left(\frac{-1-\sqrt{-3}}{2}\right)^7$ (ii) $(-1+\sqrt{-3})^5 + (-1-\sqrt{-3})^5$

4. Show that: $(1-\omega+\omega^2)(1-\omega^2+\omega^4)(1-\omega^4+\omega^8)(1-\omega^8+\omega^{16})\dots 2n \text{ factors} = 2^{2n}$

5. Prove that: $\left(\frac{i+\sqrt{3}}{2}\right)^8 + \left(\frac{i-\sqrt{3}}{2}\right)^8 = -1$

6. Evaluate: $\sum_{k=0}^5 \omega^{2k}$, where ω is an imaginary cube root of unity.

LONG QUESTIONS

1. If ω is an imaginary cube root of unity, prove that $\frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c} = \omega$

2. If ω is an imaginary cube root of unity, prove that $\frac{a\omega^{12} + b\omega^{17} + c\omega^{19}}{a\omega^{14} + b\omega^{22} + c\omega^{30}} = \omega$

MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 4

MCQs

(1) The cofactor of an element a_{ij} denoted by A_{ij} is:

- (a) $(-1)^{i-j} M_{ij}$ (b) $(-1)^{i+j} M_{ij}$ (c) $(-1)^{i+j} A_{ij}$ (d) $(-1)^{ij} M_{ij}$

(2) $(A^{-1})^t$ is equal to:

- (a) A (b) $A^{-1}A^t$ (c) $(A^t)^{-1}$ (d) $-A^t$

(3) If A is a matrix of order 3×4 then order of AA^t is:

- (a) 4×3 (b) 3×4 (c) 4×4 (d) 3×3

(4) If $\begin{vmatrix} k & 4 \\ 4 & k \end{vmatrix} = 0$, then $k =$:

- (a) 16 (b) 0 (c) ± 4 (d) 8

(5) The inverse of a square matrix exists if A is:

- (a) Singular (b) Non-Singular (c) Symmetric (d) Rectangular

(6) If A and B are matrices then $(AB)^t$ is equal to:

- (a) $B^t A^t$ (b) $A^t B^t$ (c) AB (d) BA

(7) A square matrix A is skew symmetric if $A^t =$

- (a) $-A$ (b) A (c) $\bar{A}$ (d) A^t

(8) If order of matrix is $m \times n$ then order of A^t is:

- (a) $m \times n$ (b) $m \times m$ (c) $n \times m$ (d) $n \times n$

(9) If matrix $\begin{bmatrix} x & 4 \\ 2 & 8 \end{bmatrix}$ is singular then $x =$:

- (a) 0 (b) -1 (c) 2 (d) 1

(10) If $\begin{bmatrix} x-3 & 1 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -5 & -4 \end{bmatrix}$, then $x =$:

- (a) 5 (b) -5 (c) -1 (d) 1

(11) If $A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 5 \\ 6 & 7 & 3 \end{bmatrix}$, then $A_{33} =$:

- (a) -1 (b) 1 (c) 3 (d) 0

(12) If $A = [a_{ij}]_{2 \times 2}$, then $|kA| =$:

- (a) $|A|$ (b) $k^2 |A|$ (c) $k |A|$ (d) $k |A|^2$

(13) A matrix of order $1 \times n$ is called:

- (a) Unit matrix (b) Diagonal matrix (c) Column matrix (d) Row matrix

(14) If $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 2$, then $\begin{vmatrix} b & a \\ d & c \end{vmatrix} =$:

- (a) 2 (b) -2 (c) ± 2 (d) 0

(15) Let 'A' be a square matrix of order 3×3 , and $|A| = 2$, then $|2A|$ is equal to:

- (a) 12 (b) 16 (c) 8 (d) 4

EXERCISE 4.1

SHORT QUESTIONS

1. Define:

- (i) Diagonal matrix and give one example.

(ii) Scalar matrix and give one example.

(iii) Multiplication of matrices

(iv) Transpose of a Matrix

2. If $A = [a_{ij}]_{3 \times 4}$, then show that $AI_4 = A$

3. If A and B are square matrices of the same order, then explain why in general:

(i) $(A - B)^2 \neq A^2 - 2AB + B^2$

(ii) $(A + B)(A - B) \neq A^2 - B^2$

4. If $A = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \\ -3 & 5 & 3 \end{bmatrix}$, then find $A + A'$, $A - A'$ and $(A')'$.

5. If A and B are two matrices such that $AB = B$ and $BA = A$, show that $A^2 + B^2 = A + B$.

EXERCISE 4.2

SHORT QUESTIONS

1. Define minor and cofactor of an element of a matrix.

2. If all the entries of a column of a square matrix A are zero, then show that $|A| = 0$.

3. Write any two properties of determinant.

4. Evaluate the following determinants:

(i)
$$\begin{vmatrix} 1 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 3 & 2 \end{vmatrix}$$

(ii)
$$\begin{vmatrix} a+b & a-b & a \\ a & a+b & a-b \\ a-b & a & a+b \end{vmatrix}$$

5. Without expansion, show that:

(i)
$$\begin{vmatrix} 7 & 8 & 9 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

(ii)
$$\begin{vmatrix} -a & 0 & b \\ 0 & a & -c \\ c & -b & 0 \end{vmatrix} = 0$$

(iii)
$$\begin{vmatrix} l & m+n & 1 \\ m & n+l & 1 \\ n & l+m & 1 \end{vmatrix} = 0$$

(iv)
$$\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

6. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -5 & 0 \\ -2 & -2 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & -2 & 5 \\ -3 & -1 & 4 \\ -2 & -1 & 2 \end{bmatrix}$, then find:

(i) A_{13}, A_{33}

(ii) B_{31}, B_{32}, B_{33} and $|B|$

7. Find the value of x if:

(i) $\begin{vmatrix} 2 & 1 & x \\ -1 & -4 & -3 \\ x & 1 & 0 \end{vmatrix} = 5$

(ii) $\begin{vmatrix} 1 & x-1 & 3 \\ -1 & x+1 & 2 \\ 2 & -3 & x \end{vmatrix} = 9$

8. Find the value of λ if $A = \begin{bmatrix} 4 & 2 & 3 \\ 7 & \lambda & 6 \\ 2 & 3 & 1 \end{bmatrix}$ is a singular matrix.

9. Verify that $(AB)^t = B^t A^t$ if $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ -3 & -2 \\ 0 & 1 \end{bmatrix}$

LONG QUESTIONS

1. Use properties of determinants, show that:

(i)
$$\begin{vmatrix} a+x & a & a \\ a & a+x & a \\ a & a & a+x \end{vmatrix} = x^2(3a+x)$$

(ii)
$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

(iii)
$$\begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix} = 4a^2b^2c^2$$

$$(iv) \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$$

$$(v) \begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x+y+z)(x-y)(y-z)(z-x)$$

$$(vi) \begin{vmatrix} 1 & 1 & 1 \\ a^2+1 & b^2+1 & c^2+1 \\ a^3+a & b^3+b & c^3+c \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca-1)$$

(vii)
$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ac + bc + ca$$

(viii)
$$\begin{vmatrix} 2x+3 & x+2 & x+a \\ 2x+5 & x+3 & x+b \\ 2x+7 & x+4 & x+c \end{vmatrix} = 0, \text{ where } a, b, c \text{ are in A.P.}$$

EXERCISE 4.3

LONG QUESTIONS

1. Solve the following systems of linear equations by Cramer's rule:

$$2x + y - z = 1$$

(i) $x - y + 2z = 3$

$$3x + 2y + z = 4$$

$$2x_1 - x_2 + x_3 = 1$$

(ii) $x_1 + 2x_2 + 2x_3 = 2$

$$x_1 - 2x_2 - x_3 = 1$$

2. Solve the following systems of linear equations by matrix inversion method:

$$x - 2y + z = -1$$

(i) $3x + y - 2z = 4$

$$y - z = 1$$

$$2x_1 + x_2 + 3x_3 = 3$$

(ii) $x_1 + 3x_2 - 2x_3 = 0$

$$-3x_1 - x_2 + 2x_3 = 4$$

MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 5

MCQs

(1) $x^2 - x - 2 = (x+1)(x-2)$ is:

- (A) Quadratic equation (B) Conditional equation (C) Linear equation (D) Identity

(2) $\frac{x^2}{(x^2-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}$

- (A) $\frac{C}{x+1}$ (B) $\frac{C}{x-1}$ (C) $\frac{C}{(x+1)^2}$ (D) $\frac{x+1}{x^2+1}$

(3) Partial fraction of $\frac{1}{x^2-1}$ will be of form:

- (A) $\frac{Ax+B}{x^2-1}$ (B) $\frac{A}{x+1} + \frac{B}{x-1}$ (C) $\frac{A}{x+1}$ (D) $\frac{B}{x-1}$

(4) The fraction $\frac{x^2+7x+3}{x+1}$ is:

- (A) Improper (B) Proper (C) Equivalent (D) Identity

(5) Partial fractions of $\frac{1}{(x^2+1)(x-1)}$ are of form:

- (A) $\frac{A}{x^2+1} + \frac{B}{x-1}$ (B) $\frac{A}{x+1} + \frac{B}{x^2+1} + \frac{C}{x-1}$ (C) $\frac{A}{x^2+1} + \frac{Bx+C}{x-1}$ (D) $\frac{Ax+B}{x^2+1} + \frac{C}{x-1}$

(6) From the identity $5x+4 = A(x-1) + B(x+2)$, then value of B =:

- (A) -3 (B) 3 (C) -2 (D) 2

(7) Partial fractions of $\frac{1}{x^3-1}$ will be of the form:

- (A) $\frac{A}{x+1} + \frac{Bx+C}{x^2+x+1}$ (B) $\frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$ (C) $\frac{A}{x-1} + \frac{Bx+C}{x^2-x+1}$ (D) $\frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}$

(8) The given form $(x-4)^2 = x^2 - 8x + 16$ is called:

- (A) Transidental equation (B) Cubic equation (C) An equation (D) An identity

- (9) The rational fraction $\frac{x^2}{x^2-1}$ is:
 (A) Improper (B) Proper (C) Polynomial (D) Equation
- (10) The fraction $\frac{p(x)}{x^2+1}$ will be proper rational fraction if degree of $p(x)$ is:
 (A) 4 (B) 3 (C) 2 (D) 1
- (11) Types of rational fraction are:
 (A) 1 (B) 2 (C) 3 (D) 4
- (12) Partial fractions of $\frac{x^2+1}{(x+1)(x-1)}$ will be of the form:
 (A) $\frac{A}{x+1} + \frac{B}{x-1}$ (B) $1 - \frac{A}{x+1} - \frac{B}{x-1}$ (C) $1 + \frac{A}{x+1} + \frac{B}{x-1}$ (D) $\frac{Ax+B}{x+1} + \frac{C}{x-1}$

EXERCISE 5.1

SHORT QUESTIONS

1. Define

(i) partial fraction with example.

(ii) rational fraction with example.

(iii) proper rational fraction with example.

(iv) improper rational fraction with example.

(v) an identity with example.

(vi) conditional equation with example.

2. Resolve $\frac{7x+25}{(x+3)(x+4)}$ into partial fractions.

3. Resolve $\frac{2}{x^2-1}$ into partial fractions.

4. Resolve $\frac{x^2+1}{(x+1)(x-1)}$ into partial fractions.

5. Resolve $\frac{x+1}{(x-1)^2}$ into partial fractions.

[illegible]

6. Resolve $\frac{x^2+x}{(x^2-1)^2}$ into partial fraction without determining the constants.

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7. Change $\frac{2x^3 + x^2 - x - 3}{x(2x+3)(x-1)}$ in to proper fraction.

[illegible]

LONG QUESTIONS

1. Resolve $\frac{x^2+4x+5}{(x+1)(x^2+5x+6)}$ into partial fractions.

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2. Resolve $\frac{3x^2 + 4x - 5}{(x-1)^3}$ into partial fractions.

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3. Resolve $\frac{4x^2 - 3x + 1}{(x+1)(x-1)^2}$ into partial fractions.

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4. Resolve $\frac{12x^2 - 48}{(x-2)^2(x+2)^2}$ into partial fractions.

This image shows a blank sheet of white paper with horizontal blue or grey ruling lines. At the top center, there is a red arrow pointing downwards. The paper appears to be part of a notebook or a document template.

EXERCISE 5.2

SHORT QUESTIONS

1. Resolve $\frac{2x+1}{(x-2)(x^2+3x+5)}$ into partial fractions without finding the values of unknown constants.

2. Resolve $\frac{4x^2}{(x^2+1)^2(x-1)}$ into partial fractions without finding the values of unknown constants.

[illegible]

LONG QUESTIONS

1. Resolve $\frac{2x^2 + 3x + 3}{(x+1)(x^2 + 1)}$ into partial fractions.

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2. Resolve $\frac{3x^2+3}{x^3+1}$ into partial fractions.

[illegible]

3. Resolve $\frac{2x+32}{(x-2)(x^2+2)^2}$ into partial fractions.

MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 9

MCQS

- (1) Degree of a constant polynomial is:
(A) 1 (B) 0 (C) 2 (D) Arbitrary
- (2) $P(x) = \sqrt{2}x^3 + 5x(x^2 + x^3) + 3x + \sqrt{7}$ is:
(A) Polynomial with degree 3 and leading coefficients $\sqrt{2}$
(B) Polynomial with degree 4 and leading coefficients 5
(C) Polynomial with degree 5 and leading coefficients 4
(D) Not a polynomial
- (3) What is the degree of $x^3 - 2x^4y^2 + 7y^5$ is:
(A) 6 (B) 5 (C) 4 (D) 3
- (4) Given the polynomial $P(x) = 3x^2 - 4x + 5$, what is the value of $p(2)$?
(A) 5 (B) 11 (C) 15 (D) 9
- (5) When $9x^2 - 6x + 2$ is divided by $x - 3$, the remainder will be:
(A) 60 (B) $\frac{15}{2}$ (C) $\frac{19}{5}$ (D) 65
- (6) If $P(x)$ is divided by $(x - r)$ such that $Q(x)$ is the quotient and R is the remainder then the division algorithm:
(A) $P(x) = (x - r)Q(x) + R$ (B) $P(x) = (x - r)R + Q(x)$
(C) $Q(x) = (x - r)P(x) + R$ (D) $Q(x) = (x - r)R + P(x)$
- (7) Remainder is 11 if $x^2 + 3x + 7$ is divided by:
(A) $x + 1$ (B) $x + 2$ (C) $x + 3$ (D) $x - 1$
- (8) If $x - a$ is a factor of polynomial $f(x)$, then $f(a)$ is:
(A) = 0 (B) < 0 (C) > 0 (D) $\neq 0$
- (9) If $x - 2$ is a factor of polynomial $x^3 + 2x^2 + kx + 4$ then k equals:
(A) 10 (B) -10 (C) 2 (D) 4
- (10) If the polynomial $x^3 + 9x^2 + 23x + 15$ is divided by $x + 1$, then the depressed equations is:
(A) $x^2 + 8x + 15 = 0$ (B) $x^2 - 8x + 15 = 0$ (C) $x^2 + 8x - 15 = 0$ (D) $x^2 - 8x - 15 = 0$

- (11) Suppose a polynomial regression model $P(x) = x^2 + 2x + 1$. What is the predicted value at $x = 2$?
 (A) 5 (B) 9 (C) 7 (D) 3
- (12) If the actual value is 10 and the predicted value is 8, what is the percentage error?
 (A) 10 (B) 15 (C) 20 (D) 25
- (13) Consider a data set of monthly sales figure. A polynomial regression model $P(x) = x^3 + x^2 + 2x + 1$ is fitted to this data. If the observed sales in the 3rd month are 40 units. Find the percentage error:
 (A) 7.5 (B) 8.5 (C) 9.5 (D) 10.5
- (14) A digital signal processing system has a transfer function with a numerator $B(z) = z^2 - z - 2$. Find the zeros of the system:
 (A) $z = 1, z = 2$ (B) $z = -1, z = -2$ (C) $z = 1, z = -2$ (D) $z = -1, z = 2$
- (15) For a transfer function $H(z) = \frac{(z+8)(z+14)}{z(z+4)(z+10)}$ the poles are located at:
 (A) $z = -8, z = -14$ (B) $z = 0, z = -4, z = -10$
 (C) $z = 0, z = 4, z = 10$ (D) $z = 8, z = 14$

EXERCISE 9.1

SHORT QUESTIONS

1. State the remainder and factor theorem.

2. Find remainder and quotient by simplifying the following:

(i) $(x^3 + 12x^2 - 3x + 4) \div (x - 2)$

(ii) $(5x^4 - 3x^2 + 2x^2 - 1) \div (x^2 + 4)$

3. Use the remainder theorem to find the remainder when the first polynomial is divided by the second polynomial:

(i) $x^3 + 5x^2 + 6, x + 1$

(ii) $x^4 + x^3 + 2, x + 2$

4. Use the factor theorem to determine the first polynomial is a factor of the second polynomial:

(i) $x + 1, x^3 + x^2 + x - 3$

(ii) $x - 3, x^4 - 3x^3 + x^2 - x + 1$

5. Use synthetic method to show that x is the zero of the polynomial and use the result to factorize the polynomial completely:

(i) $x^3 - 7x + 6, x = 2$

(ii) $2x^4 + 7x^3 - 4x^2 - 27x - 18, x = 2, x = -3$

6. When the polynomial $4x^4 + 2x^3 + kx^2 + 13$ is divided by $x + 1$, the remainder is 16. Find the value of k .
-
-
-

LONG QUESTIONS

1. If $x+1$ and $x-2$ are factors of $x^3 - px^2 + qx + 2$. Using synthetic division, find the values of p and q .

2. Use factor theorem to find the values a and b if -2 and 2 are roots of the polynomial $2x^3 + 4x^2 + ax + b$.

EXERCISE 9.2

SHORT QUESTIONS

1. Consider a data set at monthly sales figure. A polynomial regression model $P(x) = x^3 + 2x^2 + x - 3$ is fitted to this data. If the observed sale in the 5th month is 240 units, find the percentage error.

2. A digital signal process system has a transfer function with a nominator $B(z) = z^2 - z - 2$. Use factor theorem to find the zeros of the system.

3. A digital system has a transfer function $H(z) = \frac{z^2 + 3z + 2}{z^2 - 0.2z + 0.9}$. Find the zero(s) of the transfer function by using factor theorem.

4. A signal processing system has a transfer function with a denominator $A(z) = z^2 - 0.3z - 0.4$. Use factor theorem to find the poles of the system and determine if the system is stable.

5. The denominator of signal processing system's transfer function equal to $A(z) = z^2 + 1.2z + 0.35$. Use factor theorem to determine the locations of the corresponding poles and assess the stability of the system.

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MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 12

MCQs

1. If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial function then $\lim_{x \rightarrow c} P(x) =$:

- (A) $P(c)$ (B) 0 (C) 1 (D) $P(x)$

2. $\lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5}) =$:

- (A) -1 (B) 0 (C) 2 (D) -2

3. If k is any real number, then $\lim_{x \rightarrow a} [kf(x)] =$:

- (A) $\lim_{x \rightarrow a} k + \lim_{x \rightarrow a} f(x)$ (B) $\lim_{x \rightarrow a} [f(x)]^{\frac{1}{2}}$
(C) $k \cdot \lim_{x \rightarrow a} f(x)$ (D) $\lim_{x \rightarrow a} f(x)$

4. $\lim_{x \rightarrow 1} (2x)(x+4) =$:

- (A) 10 (B) 12 (C) 5 (D) 20

5. $\lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 + 4x^2} =$:

- (A) $\frac{1}{5}$ (B) $\frac{1}{2}$ (C) 0 (D) 4

6. $\lim_{x \rightarrow 4} \frac{x^2 - 6x + 8}{x - 4} =$:

- (A) 4 (B) 2 (C) 6 (D) 8

7. $\lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1} =$:

- (A) 0 (B) ∞ (C) -1 (D) 2

8. $\lim_{x \rightarrow a} [f(x)]^n =:$

- (A) $\lim_{x \rightarrow a} f(x)$ (B) $n \times \left[\lim_{x \rightarrow a} f(x) \right]$ (C) $\left[\lim_{x \rightarrow a} f(x) \right]^n$ (D) None of these

9. $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} =:$

- (A) na^n (B) na^{n-1} (C) 0 (D) na^{n+1}

10. $\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} =:$

- (A) $2\sqrt{2}$ (B) 0 (C) ∞ (D) $\frac{1}{2\sqrt{2}}$

11. Which of the following sequences is divergent?

- (A) $a_n = \frac{1}{2}$ (B) $a_n = (-1)^n$ (C) $a_n = \frac{n}{n+1}$ (D) $a_n = 3 - \frac{2}{n^2}$

12. The sequence $a_n = \frac{3n+1}{n+2}$ converges to:

- (A) 0 (B) 1 (C) 2 (D) 3

13. The limit of the sequence $a_n = \frac{5n+3}{2n-1}$ is:

- (A) 0 (B) $\frac{5}{2}$ (C) 3 (D) Does not exist

14. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{3n} \right)^n =:$

- (A) e^3 (B) $\frac{1}{e^3}$ (C) $e^{\frac{1}{3}}$ (D) $-\frac{1}{e^3}$

15. $\lim_{h \rightarrow 0} (1 + 2h)^{\frac{1}{h}}$ is equal to:

- (A) e^2 (B) e (C) $\frac{1}{e}$ (D) $\frac{1}{e^2}$

16. $\lim_{x \rightarrow 0} (1+3x)^{\frac{2}{x}} =:$

- (A) e^2 (B) e^8 (C) e^6 (D) e^4

17. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} =:$

- (A) $\ln a$ (B) $\ln 1$ (C) $\log_a e$ (D) $\log_e x$

18. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ is equal to:

- (A) $\ln a$ (B) 0 (C) 1 (D) ∞

19. If θ is measured in radian then $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} =:$

- (A) 1 (B) 0 (C) -1 (D) $\frac{1}{2}$

20. $\lim_{\theta \rightarrow 0} \frac{\sin 7\theta}{\theta}$ is equal to:

- (A) $\frac{1}{7}$ (B) 7 (C) 1 (D) $\frac{2}{7}$

21. $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} =:$

- (A) 0 (B) 1 (C) ∞ (D) 2

22. $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} =:$

- (A) $\frac{b}{a}$ (B) a (C) $\frac{a}{b}$ (D) $\frac{1}{b}$

23. $\lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 + \cos p\theta}$ equals:

- (A) 1 (B) 0 (C) -1 (D) 2

24. For $f(x) = \begin{cases} -2x-9, & x \leq -3 \\ 2x+3, & x > -3 \end{cases}$, $\lim_{x \rightarrow -3} f(x) =$

- (A) -3 (B) 3 (C) 6 (D) 9

25. If $f(x)$ is continuous at point $x = a$ then:

- (A) $f(a) = \lim_{x \rightarrow a} f(x)$ (B) $f(a) = \lim_{x \rightarrow 0} f(x)$
 (C) $f(0) = \lim_{x \rightarrow a} f(x)$ (D) $f(x) = \lim_{x \rightarrow a} f(x)$

26. $f(x) = 3x^2 - 5x + 4$ is continuous at $x =$:

- (A) 1 only (B) -1 only (C) 2 only (D) All real numbers

27. For what value of k , the function $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$?

- (A) 0 (B) -1 (C) -2 (D) 1

28. On which of the following intervals is $f(x) = \frac{1}{\sqrt{x-2}}$ continuous?

- (A) $[2, \infty)$ (B) $(-\infty, \infty)$ (C) $(2, \infty)$ (D) $[1, 2)$

29. The function $f(x) = \frac{2+3x}{2x}$ is not continuous at:

- (A) $x = -3$ (B) $x = -\frac{2}{3}$ (C) $x = 0$ (D) $x = 1$

30. The function $f(x) = \frac{x}{x^2-4}$ is discontinuous at:

- (A) $x = 0$ (B) $x = \pm 2$ (C) $x = 1$ (D) $x = \pm 1$

EXERCISE 12.1

SHORT QUESTIONS

1. Evaluate: $\lim_{x \rightarrow 3} \left(\frac{x-3}{\sqrt{x}-\sqrt{3}} \right)$

2.

Evaluate:

(i)

(ii)

[illegible]

3.

Evaluate:

$$\lim_{\theta \rightarrow 0} \left(\frac{1 - \cos \theta}{\theta} \right)$$

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4.

Find the limit of the following sequences if exists:

(i)

(ii)

[illegible]

5.

Find each limit by using algebraic techniques:

(i)

(ii)

[illegible]

$$(iii) \quad \lim_{x \rightarrow 2} \left(\frac{x^3 - 6x^2 + 12x - 8}{x^3 - 4x} \right)$$

$$(iv) \quad \lim_{x \rightarrow 1} \left(\frac{x^4 - 1}{x^2 - 3x + 2} \right)$$

$$(v) \quad \lim_{x \rightarrow 2} \left(\frac{x-2}{\sqrt{x+2} - \sqrt{6-x}} \right)$$

$$(vi) \quad \lim_{h \rightarrow 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \right)$$

6. Evaluate the following limits:

$$(i) \quad \lim_{x \rightarrow 0} \left(\frac{\sin x^\circ}{x} \right)$$

$$(ii) \quad \lim_{\theta \rightarrow 0} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

$$(iii) \quad \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\sin x - \cos x}{x - \frac{\pi}{4}} \right)$$

$$(iv) \quad \lim_{x \rightarrow 0} \left(\frac{\cos x - \cos bx}{x^2} \right)$$

(vii) $\lim_{x \rightarrow 2} \left(\frac{e^x - e^2}{x - 2} \right)$

LONG QUESTIONS

1. Evaluate:

(i) $\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{\cos cx - \cos dx}$

(ii) $\lim_{x \rightarrow 3} \frac{x^2 - x \log x + 3 \log x - 9}{x - 3}$

(iii) $\lim_{x \rightarrow 0} \frac{x(2^x - 1)}{1 - \cos x}$

2. Prove that:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

here

3. Prove that:

$$\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log_e a$$

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4. If θ is measured in radian, then

$$\lim_{\theta \rightarrow 0} \left(\frac{\sin \theta}{\theta} \right) = 1$$

[illegible]

EXERCISE 12.2

SHORT QUESTIONS

1. Consider the function $f(x) = \frac{x^2 - 1}{x - 1}$, discuss the continuity of f at $x = 1$.

2. Discuss the continuity of the function $f(x)$ and $g(x)$ at $x = 3$.

(i)
$$f(x) = \begin{cases} \frac{x^2 - 9}{x - 3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases}$$

(ii)
$$g(x) = \begin{cases} \frac{x^2 - 9}{x - 3} & \text{if } x \neq 3 \end{cases}$$

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1. If $f(x) = \begin{cases} 3x & \text{if } x \leq -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \geq 2 \end{cases}$. Discuss continuity of $f(x)$ at $x = 2$ and $x = -2$

1. If $f(x) = \begin{cases} 3x & \text{if } x \leq -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \geq 2 \end{cases}$. Discuss continuity of $f(x)$ at $x = 2$ and $x = -2$

2. Find the values of m and n , so that given function f is continuous at $x = 3$:

(i)
$$f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x + 9 & \text{if } x > 3 \end{cases}$$

(ii)
$$f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases}$$

3. If $f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} & \text{if } x \neq 2 \\ k & \text{if } x = 2 \end{cases}$. Find the value of k so that f is continuous at $x = 2$.

MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 13

1. The slope of the tangent line to the graph of $f(x) = x^2 - 2$ at $x = 1$ is:
(A) 2 (B) 4 (C) -2 (D) -4
2. The slope of the graph of $f(x) = 2x - 3$ at the point $(2, 1)$ is:
(A) -2 (B) -4 (C) 2 (D) 4
3. Which of the following is an equation of the tangent line to $f(x) = 9 - x^2$ at $(2, 5)$?
(A) $y = \frac{1}{4}x + \frac{9}{2}$ (B) $y = -4x + 13$ (C) $y = 4x + 13$ (D) $y = 4x - 13$
4. For which value of x is the tangent $f(x) = -x^2 - 4x - 6$ to horizontal?
(A) -8 (B) -6 (C) -4 (D) -2
5. The instantaneous rate of change of y with respect to x is given by:
(A) $\frac{\delta y}{\delta x}$ (B) $\frac{\delta x}{\delta y}$ (C) $\frac{dy}{dx}$ (D) $\frac{dx}{dy}$
6. $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ is equal to:
(A) $f'(x)$ (B) $f(x)$ (C) $f'(a)$ (D) $f(a)$
7. The instantaneous rate of $y = x^2 + 1$ change of with respect to x when $x = -4$ is:
(A) 8 (B) -8 (C) -4 (D) 4
8. The function f is given by $f(x) = 1 + 2\sin x$. What is the average rate of change of f over the interval $\left[0, \frac{\pi}{2}\right]$?
(A) $\frac{4}{\pi}$ (B) $\frac{6}{\pi}$ (C) $\frac{8}{\pi}$ (D) 2
9. Which one is Leibniz notation for the derivative of $f(x)$?
(A) $\frac{df}{dx}$ (B) $f'(x)$ (C) $f^*(x)$ (D) $Df(x)$
10. Lagrange used _____ notation for derivative:
(A) $Df(x)$ (B) $f'(x)$ (C) $\frac{dy}{dx}$ (D) $f''(x)$
11. $\frac{d}{dx}(\sin 2\pi) =$:
(A) $\cos 2\pi$ (B) $2\cos 2\pi$ (C) 0 (D) $-\cos 2\pi$

12. $\frac{d}{dx}[5f(x)] =$:
 (A) $5f'(x)$ (B) 5 (C) $f'(x)$ (D) $5 + f'(x)$

13. If $y = \frac{1}{x^2}$, then $\frac{dy}{dx}$ at $x = -1$ is:

- (A) 2 (B) 3 (C) $\frac{1}{3}$ (D) 4

14. If $y = x - \frac{1}{x}$, then $\frac{dy}{dx} =$:

- (A) $1 + \frac{1}{x^2}$ (B) $1 - \frac{1}{x^2}$ (C) $1 + \frac{1}{x}$ (D) $1 - \frac{1}{x}$

15. $\frac{d}{dx}(x-5)(3-x) =$:

- (A) $2x+8$ (B) $-2x+8$ (C) $2x-8$ (D) $x+8$

16. The derivative of $\frac{x^3 + 2x^2}{x^3}$ equals:

- (A) $\frac{2}{x^2}$ (B) $-\frac{2}{x^2}$ (C) $\frac{1}{2x^2}$ (D) $-\frac{1}{2x^2}$

17. $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] =$:

- (A) $\frac{f(x)g'(x) - f'(x)g(x)}{[g(x)]^2}$ (B) $\frac{f'(x)g(x) - f(x)g'(x)}{[f(x)]^2}$
 (C) $\frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$ (D) $\frac{g'(x)f'(x) - g(x)f(x)}{[g(x)]^2}$

18. Find $\frac{ds}{dt}$ if $s = (1+t)\sqrt{t}$

- (A) $\frac{1}{2\sqrt{t}}$ (B) $\frac{1+3t}{\sqrt{t}}$ (C) $\frac{1-3t}{2\sqrt{t}}$ (D) $\frac{1+3t}{2\sqrt{t}}$

19. $\frac{d}{dx}\left(3x^{\frac{4}{3}}\right) =$:

- (A) $4x$ (B) $4x^{\frac{1}{3}}$ (C) x^4 (D) $4x^{\frac{1}{3}}$

- [illegible]

- (i) $(3x-2)^{-2}$

[illegible]

-
-
-

- (ii) Find the instantaneous velocity at $t = 2$

6. A stone is thrown upwards and its height after t seconds is given by $S(t) = -16t^2 + 32t + 10$ (in feet). Find the instantaneous velocity at $t = 1$

7. The outdoor temperature (in $^{\circ}C$) over time is modeled by $T(t) = -t^2 + 12t + 10$, where t is the time in hours. Find the instantaneous rate of change at $t = 2$

LONG QUESTIONS

1. Find the gradient and equation of tangent line to $f(x) = 3x^2 - 4x + 1$ at $x = 2$.

2. Find the coordinates of the point of tangency and the equation of the tangent line for $f(x) = x^3 - 2x + 1$ at $x = 1$.

EXERCISE 13.2

SHORT QUESTIONS

1. Differentiate w.r.t. 'x':

(i) $x^{-3} - 2x^{-\frac{3}{2}} + 3$

(ii) $\frac{2x-3}{2x+1}$

(iii) $\left(\sqrt{x} - \frac{1}{x}\right)^2$

(iv) $\frac{x^2+1}{x^2-3}$

(v) $\frac{2x-1}{\sqrt{x^2+1}}$

(vi) $\sqrt{\frac{a-x}{a+x}}$

2. If $y = \sqrt{x} - \frac{1}{\sqrt{x}}$, show that $2x \frac{dy}{dx} + y = 2\sqrt{x}$

3. If $y = x^4 + 2x^2 + 2$, prove that $\frac{dy}{dx} = 4x\sqrt{y-1}$

LONG QUESTIONS

1. Differentiate $\frac{\sqrt{x+1}}{\sqrt{x^2-1}}$ w.r.t. x .

2. Find $\frac{dy}{dx}$ if $y = \frac{(\quad)}{x^{\frac{1}{2}} - 1}$, $x \neq 1$.

3. Differentiate $\frac{(\sqrt{x}+1)\left(x^{\frac{3}{2}}-1\right)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$ with respect to x .

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MATHEMATICS

F.Sc / ICS – 1st Year

Practice Sheet Chapter 14

MCQs

- (1) The angle between the vectors $2\hat{i} + 3\hat{j} + \hat{k}$ and $2\hat{i} - \hat{j} - \hat{k}$ is:
(A) 90° (B) 45° (C) 60° (D) 30°
- (2) $\hat{j} \times \hat{k}$ is equal to:
(A) $-\hat{i}$ (B) $\hat{i}$ (C) 0 (D) 1
- (3) If α, β, γ be the direction angles of a vector then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$ equals:
(A) -1 (B) 0 (C) 1 (D) 2
- (4) If vectors $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - 4\hat{j} + \alpha\hat{k}$ are perpendicular then $\alpha =$:
(A) 1 (B) 2 (C) 3 (D) 4
- (5) Length of the vector $2\hat{i} - \hat{j} - 2\hat{k}$ is:
(A) 2 (B) 4 (C) 3 (D) 5
- (6) The direction cosines of y-axis are:
(A) (0,1,0) (B) (1,0,0) (C) (0,0,1) (D) (0,0,0)
- (7) Projection of vector $\vec{u}$ on vector $\vec{v}$ is:
(A) $\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$ (B) $\frac{\vec{u} \cdot \vec{v}}{|\vec{u}|}$ (C) $\frac{\vec{u} \times \vec{v}}{|\vec{v}|}$ (D) $\frac{\vec{u} \times \vec{v}}{|\vec{u}|}$
- (8) If $\vec{v}$ is any vector then vector of magnitude 5 opposite to $\vec{v}$ is:
(A) $5\vec{v}$ (B) $-5\vec{v}$ (C) $5\frac{\vec{v}}{|\vec{v}|}$ (D) $-5\frac{\vec{v}}{|\vec{v}|}$
- (9) If the dot product of two vectors is zero then the vectors will be:
(A) Perpendicular (B) Parallel (C) Collinear (D) None of these

(10) The vector $\overrightarrow{PQ}$ through $P(0,5)$ and $Q(-1,-6)$ is:

- (A) $[-1,11]$ (B) $[-1,-11]$ (C) $[0,11]$ (D) $[1,1]$

(11) Magnitude of vector $\vec{u} = 2\vec{i} - 7\vec{j}$ is:

- (A) $\sqrt{53}$ (B) $\sqrt{55}$ (C) $\sqrt{48}$ (D) $\sqrt{52}$

(12) In vectors $\vec{a} \times \vec{b} =$:

- (A) $\vec{b} \times \vec{a}$ (B) $-\vec{b} \times \vec{a}$ (C) $-\vec{b}$ (D) $-\vec{a} \times \vec{b}$

(13) The unit vector in the direction of $\vec{v} = [3, -4]$

- (A) $5[3, -4]$ (B) $\frac{1}{5}[3, -4]$ (C) $\hat{i}$ (D) $\hat{j}$

(14) $\vec{i} \times \vec{j} =$:

- (A) $\vec{k}$ (B) $\vec{i}$ (C) $-\vec{k}$ (D) $\vec{j}$

(15) If 2 and 2 are x and y components of a vector, then its angle with x -axis is:

- (A) 30° (B) 60° (C) 45° (D) 90°

(16) A vector perpendicular to both vectors $\vec{a}$ and $\vec{b}$ is:

- (A) $\vec{a} \cdot \vec{b}$ (B) $\vec{a} \times \vec{b}$ (C) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$ (D) $\vec{b} \cdot \vec{a}$

(17) If $\vec{a}$ and $\vec{b}$ are parallel vectors then $\vec{a} \times \vec{b} =$:

- (A) $\vec{0}$ (B) -1 (C) 1 (D) 2

(18) Cosine of the angle between two non-zero vectors $\vec{a}$ and $\vec{b}$ is:

- (A) $\vec{a}\vec{b}$ (B) $\frac{|\vec{a}| \cdot |\vec{b}|}{\vec{a}\vec{b}}$ (C) $\frac{\vec{a}\vec{b}}{|\vec{a}| \cdot |\vec{b}|}$ (D) $\frac{\vec{a} \times \vec{b}}{|\vec{a}| |\vec{b}|}$

EXERCISE 14.1

SHORT QUESTIONS

1. Find the magnitude of the vector $\underline{v}$ and write the direction cosines of $\underline{v}$:

(i) $\underline{v} = -4\underline{i} + 4\underline{j} + 2\underline{k}$

(ii) $\underline{v} = -6\underline{i} + 8\underline{j}$

2. Find t , so that $|2\underline{i} + (t-1)\underline{j} + t\underline{k}| = \sqrt{13}$.

3. Find a unit vector in the direction of $\underline{v} = -\underline{i} + 4\underline{j} - 8\underline{k}$.

4. If $\underline{u} = 2\underline{i} + \underline{j} - 3\underline{k}$, $\underline{v} = -\underline{i} + 4\underline{j} + 2\underline{k}$ and $\underline{w} = 3\underline{i} - 2\underline{j} + \underline{k}$. Find a unit vector parallel to $4\underline{u} - 3\underline{v} + 2\underline{w}$.

5. Find a vector whose magnitude is 5 and is parallel to $3\underline{i} + 4\underline{j} - \underline{k}$

6. The position vectors of the points A, B, C and D are $\underline{u} = \underline{i} + 2\underline{j} + \underline{k}$, $\underline{v} = 7\underline{i} + 8\underline{j} + 4\underline{k}$, $\underline{w} = -\underline{i} + \underline{k}$ and $\underline{z} = \underline{i} + 2\underline{j} + 2\underline{k}$ respectively. Show that $\overline{AB}$ is parallel to $\overline{CD}$.

7. Find the direction cosines for the given vector:

(i) $\underline{v} = 4\underline{i} + 2\underline{j} - 5\underline{k}$

(ii) $\overline{PQ}$, where $P(9, 3, 13)$ and $Q(11, 6, 19)$

EXERCISE 14.2

SHORT QUESTIONS

1. Find the cosine of the angle θ between $\underline{u}$ and $\underline{v}$:

(i) $\underline{u} = 2\underline{i} + 3\underline{j} + \underline{k}$, $\underline{v} = -\underline{i} + 2\underline{j} + 2\underline{k}$

(ii) $\underline{u} = [-3, 2, 5]$, $\underline{v} = [1, 6, -2]$

2. If $\underline{a} + \underline{b} + \underline{c} = \underline{0}$, $|\underline{a}| = 3$, $|\underline{b}| = 5$ and $|\underline{c}| = 7$. Find the angle between $\underline{a}$ and $\underline{b}$.

3. Calculate the projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$ when $\underline{a} = 2\underline{i} + 3\underline{j} - \underline{k}$,
 $\underline{b} = \underline{i} - 2\underline{j} + 4\underline{k}$

4. Find a real number α so that the vectors $\underline{u}$ and $\underline{v}$ are perpendicular $\underline{v} = \alpha\underline{i} + 2\alpha\underline{j} - \underline{k}$,
 $\underline{u} = \underline{i} + \alpha\underline{j} + 3\underline{k}$

5. Find the number z so that the triangle with vertices $A(3, 0, -2)$, $B(0, 3, 1)$ and $C(1, 1, z)$ is a right triangle with right at C .

6. Show that the vectors $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ form a right triangle.
7. Find the work done, if the point at which the constant force $\underline{F} = 2\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ is applied to an object, moves from $P_1(2, -3, 1)$ to $P_2(7, 5, 3)$.
8. A particle is displaced from the point $A(5, -5, -7)$ to $B(6, 2, -2)$ under the action of constant forces defined by $10\mathbf{i} - \mathbf{j} + 11\mathbf{k}$, $4\mathbf{i} + 5\mathbf{j} + 9\mathbf{k}$ and $-2\mathbf{i} + \mathbf{j} - 9\mathbf{k}$. Show that the total work done by the force is 102 units.
9. A force of magnitude 6 units acting parallel to $4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ displace the point of application from $A(2, -1, 3)$ to $B(7, 3, 2)$. Find the work done.

LONG QUESTIONS

1. Prove that: $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

EXERCISE 14.3

SHORT QUESTIONS

1. Compute the cross product $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$. Check your answer by showing that each $\underline{a}$ and $\underline{b}$ are perpendicular to $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$:

(i) $\underline{a} = 2\underline{i} + \underline{j} - \underline{k}$, $\underline{b} = \underline{i} + \underline{j} - \underline{k}$

(ii) $\underline{a} = -4\underline{i} + \underline{j} - 2\underline{k}$, $\underline{b} = 2\underline{i} + \underline{j} + \underline{k}$

2. Find the area of the triangle formed by the points P, Q and R : $P(2,3,5)$, $Q(1,2,0)$, $R(4,1,2)$

3. Find the area of the parallelogram whose vertices are $A(1,1,1)$, $B(4,2,3)$, $C(5,6,7)$, $D(2,5,5)$

4. If the cross product of the vectors $\underline{u} = 7\underline{i} - 4\underline{j} + 5\underline{k}$ and $\underline{v} = a\underline{i} - b\underline{j} + 3\underline{k}$ is zero, then find the values of a and b .

5. Use the definition of cross product, for any vectors $\underline{u}$, $\underline{v}$, $\underline{w}$ and scalar k , prove that:

(i) $\underline{u} \times (-\underline{u}) = \underline{0}$

(ii) $\underline{u} \times \underline{v} = -\underline{v} \times \underline{u}$

(iii) $\underline{u} \times (\underline{v} + \underline{w}) = \underline{u} \times \underline{v} + \underline{u} \times \underline{w}$

6. Prove that: $\underline{a} \times (\underline{b} + \underline{c}) + \underline{b} \times (\underline{c} + \underline{a}) + \underline{c} \times (\underline{a} + \underline{b}) = \underline{0}$

7. Use the definition of cross product, prove that for any vectors $\underline{u}$ and $\underline{v}$: $(\underline{u} + \underline{v}) \times (\underline{u} - \underline{v}) = -2(\underline{u} \times \underline{v})$

8. Find the moment about the point $M(1, -3, 3)$ of the force represented by $\overrightarrow{AB}$, where coordinates of points A and B are $A(4, 3, -1)$ and $B(-1, 3, 7)$ respectively.

9. A force $\underline{F} = -2\underline{i} + \underline{j} - 3\underline{k}$ is applied at the point $P(-1, -3, 2)$. Find its moment about the point $Q(4, 2, 2)$.

LONG QUESTIONS

1. Find a unit vector perpendicular to the plane containing $\underline{a}$ and $\underline{b}$. Also find sine of the angle between them:

(i) $\underline{a} = \underline{i} + 6\underline{j} - 3\underline{k}$, $\underline{b} = 2\underline{i} + \underline{j} + 3\underline{k}$

(ii) $\underline{a} = 5\underline{i} + \underline{j} - 3\underline{k}$, $\underline{b} = -2\underline{i} + 4\underline{j} + \underline{k}$

2. Prove that: $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

3. Show that: $|\underline{a} \times \underline{b}|^2 = |\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2$